Midterm 1 — Study Guide & Training Plan

Stats 280, Part 1 (Lectures 1–8: study design through regression). Exam: Wednesday, October 14.

How the real exam is built

W25 and W26 used the same skeleton. Expect it again:

Q	Pts	Topic	What they ask
1	30	Study design	Confounders, bias direction, observational vs. experiment, adherers/refusers, cross-tab sample size
2	30	Averages & lists	Weighted averages / Simpson's paradox, rebuild a list from percentiles, SD bounds
3	35	Histograms	Density scale, median/mean/IQR from blocks, "game" = median (abs) or mean (squares)
4	35	Normal, Markov, Chebyshev	Areas, percentiles, maximum possible percentages
5	30	Correlation	$\overline{xy}$ algebra, combined data sets, missing value for $r = 0$ or ± 1, scatter matching
6	40	Regression	Prediction, equation, RMS error, percentiles in a vertical strip, percentile → percentile

Formulas to know cold

- Percentile (class definition): the smallest number *q in the list* with at least X% of the list $\leq q$. With n numbers, need $\lceil Xn/100 \rceil$ numbers. Median = 50th percentile (the smaller middle number when n is even).
- $\text{Var} = \overline{x^2} - \bar{x}^2$. So $\overline{x^2} = SD^2 + \bar{x}^2$. This is the tool for combining groups and fixing a wrong entry.
- $\text{cov} = \overline{xy} - \bar{x}\bar{y}$; $r = \text{cov}/(SD_x SD_y)$; $\sum xy = n(r\, SD_x SD_y + \bar{x}\bar{y})$.
- $SD = \sqrt{(n-1)/n}\; SD^+$ (R's `sd()` gives SD^+).
- Markov (non-negative lists): fraction $\geq x$ is at most $\bar{x}/x$. For the lower tail of a bounded list, flip it: apply Markov to $\max - x$.
- Chebyshev: fraction k or more SDs from the mean is at most $1/k^2$. Uneven interval: use the side closer to the mean.
- Regression: slope $= r\, SD_y/SD_x$, through $(\bar{x}, \bar{y})$. Predicted $z_y = r\, z_x$. RMS $= \sqrt{1-r^2}\, SD_y$.
- Two regression lines: product of slopes $= r^2$; both pass through the point of averages.

- Normal table z's: 50% middle 0.67 · 60% 0.84 · 70% 1.04 · 80% 1.28 · 90% 1.65 · 95% 1.96. The table gives area *between* $-z$ and z; a tail is (100 − area)/2.

Traps that cost points

- Using "average of the two middle numbers" for the median. The class definition picks a number from the list.
- Using SD(y) instead of the RMS error inside a vertical strip (W25 6c, W26 6c–d).
- Percentile questions in regression: percentile → z → multiply by r → back to units, *then* use the RMS error for spread.
- Solving the y-on-x line for x. Predicting x needs the other line.
- Calling the regression effect "real" (tutoring, "easier for weak students"). It is the regression fallacy.
- Markov needs non-negative values and only bounds the upper tail directly.
- Comparing adherers to the whole control group (Clofibrate, HIP screening). Compare the randomized groups as assigned.
- Bias direction: say *which way* and *why*, naming the confounder and both of its links (to treatment and to outcome).
- Scatter plots with unequal axis scales. Rescale in your head so the SDs match.
- Change of scale: adding shifts the mean and median only; multiplying by c scales the SD and IQR by $|c|$; r flips sign only if exactly one variable gets a negative multiplier.

Training plan

- **Monday night:** Practice Midterm 1A under real conditions: 80 minutes, closed book, calculator, normal table only. Grade it with the solutions. Write down every point lost and the reason.
- **Tuesday:** Practice Midterm 1B (timed) in the morning and 1C (timed) in the evening. Grade each one right after. Redo every missed part from a blank page without looking.
- **Tuesday, last hour:** Reread this sheet and your list of lost points. Rework W25/W26 Q6 (regression) once more; it is worth 40 points.
- **Wednesday:** Skim the formulas and traps only. On the exam, write every step; unjustified answers get zero.

Target pace: about 1 point per 25 seconds, so roughly 12 minutes for each 30–35 point question and 16 for Q6. If you stall on a part for more than 3 minutes, move on.